

European Operational Program for Exchange of Weather Radar Information (OPERA)

OPERA production data and software release 2025b

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Project: OPERA Production release 2025

Production Centre: DWD, GeoSphere Austria, KNMI and Météo-France

Application Domain: CUMULUS, STRATUS, CIRRUS and NIMBUS production

Processing Software: CIRRUS 1.50, BALTRAD version dated 2025-10-01, NIMBUS

Reference	Date	Author(s)	Content
OPERA_release_notes_20251118	19.11.2025	Annakaisa von Lerber	First draft

1. OPERA production data release 2025b

1. Data exchange update

- 2 new radars new sites in Lithuania (EYHM40 & EYHM41)
- 4 new radars existing sites (LEMM42, LEMM45, LEMM50 & LEMM51) & 1 new radar new site in Spain (LEMM61)
- 1 new radar new site in Finland (EFKL55)
- 1 new radar new site in Denmark (EKMI46)
- 2 new radars new sites in Portugal (LPMG46 & LPMG47) - located outside of OPERA composite domain
- Israeli data flow (LLBD40) is cut, as IMS is no longer Member of OPERA

2. Metadata and ODIM update for all production lines

- Introduction of new scanning strategy for 2 radars in Czech Republic (OPKR50 & OPKR60), update to ODIM_H5 v2.4.2 (to be published).
- Introduction of corrected time metadata for 1 radar in Latvia (EVRR40) - reinsertion of radar in composite
- Update of radar coordinates for 12 radars in Sweden (ESWI), update to ODIM_H5 v2.4.
- Update of radar coordinates for 1 radar in Ireland (EIDB40)
- Insertion of product related metadata in HDF5 output file: license and the attributes for purposes of metadata discovery
- Clarified that WSI changes in metadata do not break the production. Instructions for Members to change WSI in the metadata will follow later.
- Check and harmonize /dataset1/what/product for both CIRRUS and NIMBUS

2. OPERA production software release 2025b

1. CUMULUS and STRATUS

- No changes

2. CIRRUS 1.5.0

- **Metadata and ODIM update in CIRRUS**
 - Update of radar coordinates for 1 radar in Norway (ENMI37) (CIRRUS only)
 - Bugfix: insertion of scans O and P for 3 radars in Portugal (LPMG41, LPMG42 & LPMG43) (CIRRUS only)
- **CIRRUS product update**
 - Improvement of compositing method: search distance for contributing radar pixels now varies with distance from radar
- **Update of satellite filter**
 - Introduction of 3rd option MTG 10-min for satellite data to ensure transition MSG to MTG

- Bugfix: when the satellite products are missing, incoming reflectivity files prone to satellite filter are not passed to the compositing
- **Update of advection task**
 - Introduction of reflectivity data from EVRR40 and EIDB40

3. NIMBUS

- **Metadata and ODIM update in NIMBUS**
 - Name change for Hungarian radar data
 - Following the new naming convention for the dual-prf scans not meant for compositing, using Y as the A1 identifier. Example name: PAHZ41.HABP --> PAYZ41.HABP
- **NIMBUS production update**
 - New BALTRAD features updated in NIMBUS
 - Change HAC-filter quality field.
 - Upgrade to Rocky Linux 9.6, and Python 3.9.
 - Updated Java to version 21
 - Updated Baltrad to version (2025-10-01) in <https://rpm.baltrad.eu/RockyLinux/9/latest/>
 - In solar monitoring, the use of default pulsewidth value is corrected.
- **Bugfixes**
 - Corner coordinates fixed in post-processing.
 - Security leak in Apache Tomcat web server.
 - Occasionally corrupt NIMBUS rain rate files.
- **Post-processing and operational services**
 - Extend rolling archive time on EWC-ftp for all products
 - Bugfix and improvements on product monitoring

4. Open Radar Data API

- Not yet involved in the OPERA production release. Included in the 2026a.

Contact

OPERA support: support.opera@eumetnet.eu

Supportive documents: